

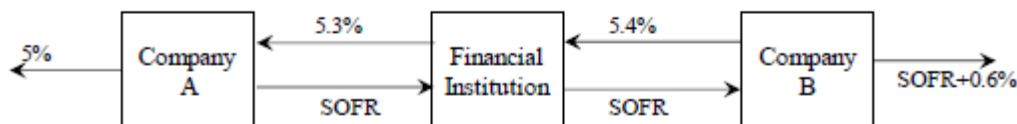
FIN302

Tutorial 9 Answers

Chapter 7

7.11

A has an apparent comparative advantage in fixed-rate markets but wants to borrow floating. B has an apparent comparative advantage in floating-rate markets but wants to borrow fixed. This provides the basis for the swap. There is a 1.4% per annum differential between the fixed rates offered to the two companies and a 0.5% per annum differential between the floating rates offered to the two companies. The total gain to all parties from the swap is therefore $1.4 - 0.5 = 0.9\%$ per annum. Because the bank gets 0.1% per annum of this gain, the swap should make each of A and B 0.4% per annum better off. This means that it should lead to A borrowing at $\text{SOFR} - 0.3\%$ and to B borrowing at 6.0%. The appropriate arrangement is therefore as shown in Figure S7.1.



7.12

Consider the party paying floating. The first exchange involves paying \$1.2 million and receiving \$2.0 million in four months. It has a value of $0.8e^{-0.027 \times 0.3333} = \0.7928 million. To value the second forward contract, we note that the forward interest rate is 3% per annum with semiannual compounding. The value of the forward contract is

$$100 \times (0.04 \times 0.5 - 0.03 \times 0.5)e^{-0.027 \times 0.8333} = \$0.4889 \text{ million}$$

The total value of the forward contracts is therefore $\$0.7928 + \$0.4889 = \$1.2817$. This is the value of the swap to the party paying floating. For the party paying fixed, the value is $-\$1.2817$.

7.13

X has a comparative advantage in yen markets but wants to borrow dollars. Y has a comparative advantage in dollar markets but wants to borrow yen. This provides the basis for the swap. There is a 1.5% per annum differential between the yen rates and a 0.4% per annum differential between the dollar rates. The total gain to all parties from the swap is therefore $1.5 - 0.4 = 1.1\%$ per annum. The bank requires 0.5% per annum, leaving 0.3% per annum for each of X and Y. The swap should lead to X borrowing dollars at $9.6 - 0.3 = 9.3\%$ per annum

and to Y borrowing yen at $6.5 - 0.3 = 6.2\%$ per annum. The appropriate arrangement is therefore as shown in Figure S7.2. All foreign exchange risk is borne by the bank.



7.14

The swap involves exchanging the sterling interest of 20×0.10 or £2 million for the dollar interest of $30 \times 0.06 = \$1.8$ million. The principal amounts are also exchanged at the end of the life of the swap. The value of the sterling bond underlying the swap is

$$\frac{2}{(1.07)^{1/4}} + \frac{22}{(1.07)^{5/4}} = 22.182 \text{ million pounds}$$

The value of the dollar bond underlying the swap is

$$\frac{1.8}{(1.04)^{1/4}} + \frac{31.8}{(1.04)^{5/4}} = \$32.061 \text{ million}$$

The value of the swap to the party paying sterling is therefore

$$32.061 - (22.182 \times 1.55) = -\$2.321 \text{ million}$$

The value of the swap to the party paying dollars is \$2.321 million. The results can also be obtained by viewing the swap as a portfolio of forward contracts. The continuously compounded interest rates in sterling and dollars are 6.766% per annum and 3.922% per annum. The 3-month and 15-month forward exchange rates are $1.55e^{(0.03922-0.06766) \times 0.25} = 1.5390$ and $1.55e^{(0.03922-0.06766) \times 1.25} = 1.4959$. The values of the two forward contracts corresponding to the exchange of interest for the party paying sterling are therefore

$$(1.8 - 2 \times 1.5390)e^{-0.03922 \times 0.25} = -\$1.2656 \text{ million}$$

and

$$(1.8 - 2 \times 1.4959)e^{-0.03922 \times 1.25} = -\$1.1347 \text{ million}$$

The value of the forward contract corresponding to the exchange of principals is

$$(30 - 20 \times 1.4959)e^{-0.03922 \times 1.25} = +\$0.0787 \text{ million}$$

The total value of the swap is $-\$1.2656 - \$1.1347 + \$0.0787$ million or $-\$2.322$ million (which allowing for rounding errors is the same as that given by valuing bonds).

7.15

Credit risk arises from the possibility of a default by the counterparty. Market risk arises from movements in market variables such as interest rates and exchange rates. A complication is that the credit risk in a swap is contingent on the values of market variables. For example, suppose that a company has a single bilaterally cleared swap with a counterparty. The company's has credit risk only when the value of the swap to the company is positive.

7.16

The rate is not truly fixed because, if the company's credit rating declines, it will not be able to roll over its floating rate borrowings at floating plus 150 basis points. The effective fixed borrowing rate then increases. Suppose, for example, that the treasurer's spread over floating increases from 150 basis points to 200 basis points. The borrowing rate increases from 5.2% to 5.7%.

7.19

At the end of year 3, the financial institution was due to receive \$200,000 ($= 0.5 \times 4\%$ of \$10 million) and pay \$150,000 ($= 0.5 \times 3\%$ of \$10 million). The immediate loss is therefore \$50,000. To value the remaining swap, we assume that LIBOR forward rates are realized. All forward rates are 2% per annum. The remaining cash flows are therefore valued on the assumption that the floating payment is $0.5 \times 0.02 \times 10,000,000 = \$100,000$. The fixed payment is \$200,000 and the net payment that would be received is $200,000 - 100,000 = \$100,000$. The total cost of default is therefore the cost of foregoing the following cash flows:

3 year:	\$50,000
3.5 year:	\$100,000
4 year:	\$100,000
4.5 year:	\$100,000
5 year:	\$100,000

Discounting these cash flows to year 3 at 1.8% per annum, we obtain the cost of the default as \$441,120.

7.21

Company A has a comparative advantage in the Canadian dollar fixed-rate market. Company B has a comparative advantage in the U.S. dollar floating-rate market. (This may be because of their tax positions.) However, company A wants to borrow in the U.S. dollar floating-rate market and company B wants to borrow in the Canadian dollar fixed-rate market. This gives rise to the swap opportunity.

The differential between the U.S. dollar floating rates is 0.5% per annum, and the differential between the Canadian dollar fixed rates is 1.5% per annum. The difference between the differentials is 1% per annum. The total potential gain to all parties from the swap is therefore 1% per annum, or 100 basis points. If the financial intermediary requires 50 basis points, each of A and B can be made 25 basis points better off. Thus, a swap can be designed so that it provides A with U.S. dollars at Floating + 0.25% per annum, and B with Canadian dollars at 6.25% per annum. The swap is shown in Figure S7.4.

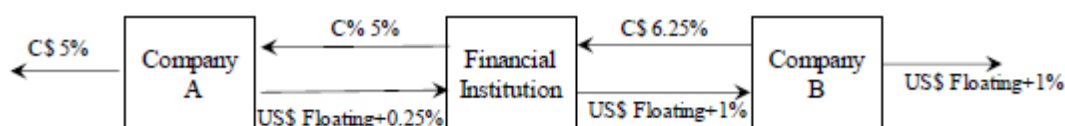


Figure S7.4 Swap for Problem 7.11

Principal payments flow in the opposite direction to the arrows at the start of the life of the swap and in the same direction as the arrows at the end of the life of the swap. The financial institution would be exposed to some foreign exchange risk which could be hedged using forward contracts.